

ReadQueue — Design Document

Project description

ReadQueue is a web application for creating and sharing curated reading lists. Users maintain separate lists of academic papers and books, track their reading progress per item, and leave comments on individual entries. Lists are shared via a unique public URL — no account or registration required. Anyone with the link can view the list and add comments; editing and adding items requires a private edit token held by the list creator.

User personas

Persona 1 — Alex Chen, PhD student

- 27 years old, second-year CS PhD student
- Runs frequent literature reviews, juggling 10–20 papers at a time
- Works in a lab of 4–5 people with a shared advisor
- **Needs:** share paper lists with lab mates, track who has read what, attach notes for co-authors on specific papers
- **Frustrations:** email threads lose context, Zotero is overkill for quick sharing, Google Docs don't track per-item reading status

Persona 2 — Sofia Martínez, book club organizer

- 34, works in UX design
- Runs an 8-person monthly book club
- **Needs:** publish next month's pick, let members mark reading progress, enable pre-meeting discussion without managing accounts for 8 people
- **Frustrations:** group chats get noisy and lose track of who is ready to discuss, members forget email announcements

Persona 3 — Marcos Rivera, graduate course TA

- 29, TA for a graduate-level seminar course
- Prepares weekly reading lists for 20 students across a semester
- **Needs:** curate readings in deliberate order, annotate each paper with focus guidance, let students ask questions on specific readings before class
- **Frustrations:** the university LMS is too clunky for per-paper discussion threads; students lose track of which readings belong to which week

User stories

Alex — reading sprint before a deadline

Alex has a paper submission in two weeks. She opens ReadQueue, creates a list called "ICML 2026 related work sprint," and adds eight papers with tags like `#transformers` and `#scaling`. She copies the share link into a Slack message to her two co-authors. Each morning she moves papers from *to-read* to *reading* to *done* and her co-authors leave comments — "skip §4, the ablation is weak," "check the follow-up by Anthropic" — directly on individual entries. No one had to sign up. By day three the list has become a shared living review document that survives across devices and sessions.

Alex — showing the advisor

Alex wants her advisor to see her reading progress before their weekly meeting. She pastes the share link into an email. Her advisor clicks it on her phone, sees every paper with its status and Alex's inline notes, and replies in the comment field. No account prompt appears. The list is still there the following week when Alex opens it from the lab desktop.

Alex — filtering by theme

Four weeks into a lit review, Alex has 30 papers on the list. She clicks the `#mechanistic-interp` tag filter and sees only the eight papers under that theme. She works through that cluster, marks them done, then switches to the `#RL` filter. The filter runs client-side — no page reload.

Sofia — setting up book club

Sofia creates a list called "June reads," adds the selected book, and writes a comment explaining her choice: "pairs well with what we discussed in May, shorter than usual." She pastes the link into the group WhatsApp. All eight members open it without signing up. Sofia does not handle a single password reset.

Sofia — checking who is ready to discuss

Two days before the meeting, Sofia opens the list. Six members have moved the book to *done*, two to *reading*. She knows to pace the meeting for mixed completion and plans two quick catch-up questions for the two who are still reading. She did not have to send a "where are you?" poll.

Marcos — weekly readings with a reading order

Marcos creates a list for Week 7. He adds four papers in deliberate order — foundational paper first, most recent last — and adds a comment on each explaining what section to focus on: "read §1–3 only for this one, the rest is not examinable." Students see the intended reading path rather than an undifferentiated pile of links.

Marcos — fielding questions before class

A student leaves a comment on paper 2: "the proof in §3 doesn't follow from the lemma — is this intentional?" Marcos sees it, replies directly in the thread, and three other students read the reply before Thursday. The question is not repeated in class. The thread stays attached to the paper for the rest of the semester.
